

Google Cloud Platform

Complete Learning Guide & Reference Notes

Concepts • Services • Architecture • Best Practices

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Introduction to Google Cloud Platform

Google Cloud Platform (GCP) is a suite of cloud computing services that runs on the same infrastructure Google uses internally for its end-user products like Google Search, Gmail, and YouTube. GCP offers services in compute, storage, networking, big data, machine learning, and the internet of things (IoT), as well as cloud management, security, and developer tools.

Key Advantages of GCP:

- Global infrastructure with 200+ countries and territories reached
- Same infrastructure as Google's consumer products
- Strong commitment to open source and multi-cloud
- Advanced data analytics and machine learning capabilities
- Competitive pricing with sustained use discounts and per-second billing
- Robust security with encryption at rest and in transit by default

Core GCP Concepts

Resource Hierarchy

GCP resources are organized hierarchically, which helps with management, billing, and access control.

Service / Concept	Description & Key Points
Organization	Root node representing your company. Optional but recommended. Allows centralized control over all GCP resources.
Folders	Grouping mechanism for projects. Can contain other folders or projects. Useful for organizing resources by department, team, or environment.
Projects	Core organizational component. All resources belong to a project. Projects are isolated from each other. Each project has unique ID, name, and number.
Resources	Individual services like VMs, buckets, databases. Always belong to a project. Inherit permissions from parent hierarchy.

Regions and Zones

GCP infrastructure is divided into regions and zones for redundancy and low latency.

Service / Concept	Description & Key Points
Regions	Independent geographic areas (e.g., us-central1, europe-west1). Each region has multiple zones. Regions are completely independent from each other. Choose regions close to users for lower latency.

Service / Concept	Description & Key Points
Zones	Deployment areas within a region (e.g., us-central1-a, us-central1-b). Single failure domains. Resources in different zones within a region have high-bandwidth, low-latency network connections.
Multi-Region	Large geographic areas containing multiple regions (e.g., US, EU, ASIA). Used for services like Cloud Storage for highest availability and redundancy.

Identity and Access Management (IAM)

IAM controls who (identity) has what access (role) to which resources. Follows principle of least privilege.

Service / Concept	Description & Key Points
Members/Principals	Who can access resources: Google Accounts (individual users), Service Accounts (applications), Google Groups, Google Workspace domains, Cloud Identity domains.
Roles	Collections of permissions. Three types: Primitive (Owner, Editor, Viewer - broad access), Predefined (specific to services, fine-grained), Custom (user-defined permissions).
Policies	Bind members to roles for resources. Can be set at organization, folder, project, or resource level. Child resources inherit policies from parents.
Service Accounts	Special accounts for applications/VMs. Used for server-to-server interactions. Can be granted roles and used for authentication. Keys can be managed or Google-managed.

Compute Services

Compute Engine

Infrastructure as a Service (IaaS) - Virtual machines running in Google's data centers.

Service / Concept	Description & Key Points
Virtual Machines	Customizable VMs with various machine types (general-purpose, compute-optimized, memory-optimized). Support for both Linux and Windows. Can attach GPUs for ML/graphics workloads.
Machine Types	Predefined (standard configurations) or Custom (specify exact vCPU and memory). E2 (cost-optimized), N2/N2D (balanced), C2/C2D (compute-intensive), M1/M2 (memory-intensive).
Persistent Disks	Durable block storage. Standard (HDD) or SSD options. Can be resized while attached. Automatically encrypted. Snapshots for backup.
Preemptible VMs	Short-lived, low-cost VMs (up to 80% discount). Can be terminated by GCP at any time. Ideal for batch jobs, fault-tolerant workloads. Max runtime 24 hours.
Spot VMs	Latest version of preemptible VMs. No max runtime. Can be reclaimed with 30-second notice. Even lower cost for interruptible workloads.
Instance Groups	Managed (identical instances from template, auto-scaling) or Unmanaged (different instances). Used with load balancers for high availability.
Sole-Tenant Nodes	Dedicated hardware for compliance/licensing requirements. Your VMs run on hardware not shared with other customers.

Google Kubernetes Engine (GKE)

Managed Kubernetes service for containerized applications with automatic scaling and updates.

Service / Concept	Description & Key Points
Cluster Types	Zonal (single zone, lower cost) or Regional (multi-zone, higher availability). Autopilot mode (fully managed) or Standard mode (more control).
Node Pools	Groups of nodes with same configuration. Can have different machine types in same cluster. Useful for workload isolation (e.g., CPU vs GPU nodes).
Autopilot Mode	Fully managed Kubernetes. Google manages nodes, scaling, security. Pay only for pod resources. Best for hands-off operation.
Standard Mode	More control over cluster configuration. Manual node management and scaling. Suitable when specific configurations needed.

Service / Concept	Description & Key Points
Workload Identity	Secure way for GKE pods to access GCP services. Maps Kubernetes service accounts to GCP service accounts. No need for key files.
Auto-Scaling	Horizontal Pod Autoscaler (scale pods based on CPU/memory). Cluster Autoscaler (add/remove nodes based on demand). Vertical Pod Autoscaler (adjust resource requests).

App Engine

Platform as a Service (PaaS) - Fully managed serverless platform for web and mobile backends.

Service / Concept	Description & Key Points
Standard Environment	Preconfigured runtimes (Python, Java, Node.js, PHP, Go, Ruby). Auto-scales to zero. Free tier available. Fast startup. Limited customization.
Flexible Environment	Runs containers. More customization. Can use any runtime. SSH access available. Doesn't scale to zero. Longer startup time.
Services	Microservices architecture support. Each service can have multiple versions. Traffic splitting for gradual rollouts and A/B testing.
Scaling	Automatic or manual scaling. Basic (based on instance count), Automatic (based on load), Manual (fixed instances). Instance hours billed.
Cron Jobs	Schedule tasks using cron.yaml. Integrated task scheduling without external services.

Cloud Functions

Function as a Service (FaaS) - Event-driven serverless compute for single-purpose functions.

Service / Concept	Description & Key Points
Event-Driven	Triggered by HTTP requests, Pub/Sub messages, Cloud Storage changes, Firestore events, Firebase events. No server management needed.
Supported Languages	Node.js, Python, Go, Java, .NET, Ruby, PHP. Each function is single-purpose and lightweight.
Scaling	Automatically scales based on demand. Each invocation runs in isolated environment. Pay only for execution time (100ms billing increments).
Use Cases	Webhooks, data processing, IoT backends, mobile backends, API endpoints, real-time file processing, scheduled tasks.
2nd Generation	Built on Cloud Run. Longer timeouts (60 min vs 9 min). Larger instances. More concurrent requests. Better event handling.

Cloud Run

Fully managed serverless platform for containerized applications with automatic scaling.

Service / Concept	Description & Key Points
Container-Based	Deploy any container that listens on a port. Use any language, library, or binary. Stateless HTTP containers. Maximum flexibility.
Fully Managed	No infrastructure management. Automatic HTTPS endpoints. Custom domains supported. Scales to zero (no cost when idle).
Cloud Run for Anthos	Run Cloud Run on GKE clusters or on-premises. Consistent developer experience. More control over underlying infrastructure.
Concurrency	Multiple requests per container instance. Configurable (1-1000). Reduces cold starts. Cost-efficient for high-traffic services.
Revisions	Immutable deployments. Traffic splitting between revisions. Easy rollbacks. Gradual rollouts.

Storage Services

Cloud Storage

Object storage for unstructured data with global accessibility and high durability.

Service / Concept	Description & Key Points
Storage Classes	Standard (hot data, frequently accessed), Nearline (once/month access, 30-day minimum), Coldline (once/quarter, 90-day minimum), Archive (once/year, 365-day minimum). Automatic transitions with lifecycle policies.
Buckets	Containers for objects. Globally unique names. Region, dual-region, or multi-region location. Versioning support. Lifecycle management.
Objects	Individual pieces of data. Max size 5 TB. Immutable (new version on update). Metadata support. ACL or IAM permissions.
Durability	11 9's (99.999999999%) annual durability. Automatic replication across zones/regions. Designed for zero data loss.
Use Cases	Website content, backups, archives, big data analytics, ML training data, content distribution, disaster recovery.
Features	Versioning, lifecycle policies, object holds, bucket lock, signed URLs, requester pays, customer-managed encryption keys (CMEK).

Persistent Disk

Block storage for Compute Engine VMs with automatic encryption and snapshots.

Service / Concept	Description & Key Points
Disk Types	Standard (HDD, cost-effective), Balanced (SSD, balance of cost/performance), SSD (high performance), Extreme (highest performance, customizable IOPS).
Zonal vs Regional	Zonal (single zone, lower cost), Regional (synchronously replicated across two zones, higher availability).
Snapshots	Point-in-time backups. Incremental (only changes backed up). Cross-region replication available. Used for disaster recovery and migration.
Resize	Can increase size while attached to running VM. Cannot decrease size. No downtime required.
Performance	Scales with disk size. Larger disks = better performance. Can attach multiple disks to single VM. Read-only disks can be shared across VMs.

Filestore

Fully managed NFS file storage for applications requiring shared file system.

Service / Concept	Description & Key Points
NFS Protocol	Industry-standard Network File System. Compatible with existing applications. Linux and Windows support.
Tiers	Basic (1-63.9 TB, cost-effective), High Scale (10-100+ TB, high throughput), Enterprise (1-10 TB, highest performance).
Use Cases	GKE persistent volumes, web serving, content management, home directories, lift-and-shift migrations, development environments.
Performance	Consistent performance. Automatic scaling. Throughput increases with capacity.

Database Services

Cloud SQL

Fully managed relational database service for MySQL, PostgreSQL, and SQL Server.

Service / Concept	Description & Key Points
Supported Engines	MySQL, PostgreSQL, SQL Server. Multiple versions supported. Easy migration from on-premises.
High Availability	Regional instances with automatic failover. Synchronous replication to standby. 99.95% SLA. Automated backups and point-in-time recovery.
Scaling	Vertical scaling (increase CPU/memory). Read replicas for read-heavy workloads. Storage auto-increase available.
Security	Automatic encryption at rest and in transit. Private IP support. Cloud SQL Proxy for secure connections. IAM integration.
Maintenance	Automatic patches and updates. Configurable maintenance windows. Minimal downtime updates.

Cloud Spanner

Globally distributed, horizontally scalable relational database with strong consistency.

Service / Concept	Description & Key Points
Global Distribution	Multi-region with synchronous replication. Strong consistency across regions. 99.999% availability SLA. External consistency (globally ordered transactions).

Service / Concept	Description & Key Points
Horizontal Scaling	Add nodes to increase throughput and storage. Linear scaling. No sharding complexity. Handles petabytes of data.
SQL Support	ANSI SQL 2011 with extensions. Supports ACID transactions. Automatic secondary indexes. Query optimizer.
Use Cases	Global applications requiring strong consistency, financial services, retail inventory, gaming leaderboards, supply chain.
Architecture	TrueTime API ensures global consistency. Automatic sharding. Data automatically re-distributed as you scale.

Firestore

NoSQL document database for mobile, web, and server development with real-time sync.

Service / Concept	Description & Key Points
Document Model	Data stored in documents (JSON-like). Documents organized in collections. Subcollections for hierarchical data. Flexible schema.
Real-Time Updates	Listen to changes in real-time. Automatic sync across clients. Offline support with local caching. Conflict resolution.
Modes	Native mode (recommended, real-time, mobile/web optimized) or Datastore mode (server workloads, backward compatible).
Querying	Powerful queries with compound indexes. Filtering and sorting. Pagination support. Real-time query results.
Scaling	Automatic horizontal scaling. Multi-region replication. Strong consistency. Serverless - no capacity planning.
Use Cases	Mobile apps, real-time collaboration, user profiles, gaming state, IoT data, social features.

Cloud Bigtable

NoSQL wide-column database for large analytical and operational workloads.

Service / Concept	Description & Key Points
Wide-Column Store	Sparse, distributed, persistent multi-dimensional sorted map. Single key lookup is fast. Row key design is critical for performance.
Performance	Sub-10ms latency. Millions of QPS. Scales linearly with nodes. HDD or SSD storage options.

Service / Concept	Description & Key Points
Use Cases	Time-series data, IoT sensor data, financial data, ad tech, recommendation engines, graph data, analytics.
Integration	HBase API compatible. Works with Hadoop, Dataflow, Dataproc. Native integration with BigQuery.
Scaling	Add nodes for more throughput. Replication for high availability. Storage automatically scales with data.

Memorystore

Fully managed in-memory data store service for Redis and Memcached.

Service / Concept	Description & Key Points
Redis	In-memory data structure store. Supports various data types (strings, hashes, lists, sets). Persistence options. Pub/Sub messaging.
Memcached	High-performance distributed memory caching. Simple key-value store. No persistence. Lower latency than Redis.
High Availability	Standard tier with automatic failover and replication. Basic tier for dev/test (single zone, no replication).
Use Cases	Session management, caching database queries, real-time analytics, leaderboards, message queues.
Performance	Sub-millisecond latency. Up to 300 GB capacity. Millions of operations per second.

Networking Services

Virtual Private Cloud (VPC)

Global private network for GCP resources with fine-grained networking controls.

Service / Concept	Description & Key Points
Global Resource	VPC spans all regions. Subnets are regional. Single VPC can have resources in multiple regions.
Subnets	Regional IP address ranges. Primary and secondary ranges. Expand without downtime. Private Google Access for accessing GCP services.
Firewall Rules	Stateful firewall. Allow/deny rules based on IP, port, protocol, tags. Hierarchical priority system. Default deny ingress, allow egress.
Routes	Define traffic paths. System routes (automatic) and custom routes. Route based on destination IP. Priority-based selection.
VPC Peering	Connect VPCs privately. Transitive peering not supported. No overlapping IP ranges. Low latency, high bandwidth.
Shared VPC	Share VPC across multiple projects in organization. Centralized network administration. Service projects use host project's network.

Cloud Load Balancing

Fully distributed, software-defined load balancing for high availability and scalability.

Service / Concept	Description & Key Points
Global Load Balancing	Single anycast IP serves traffic worldwide. Automatically routes to nearest healthy backend. Scales automatically with traffic.
HTTP(S) Load Balancing	Layer 7 load balancing. SSL/TLS termination. URL-based routing. Integration with Cloud CDN. WebSocket support.
TCP/UDP Load Balancing	Layer 4 load balancing. Network and internal load balancing. Regional or global. Preserve client IP.
Internal Load Balancing	Private load balancing within VPC. Regional resource. Load balance internal traffic. Multi-tier architectures.
Health Checks	Automatic health checking. Configurable intervals and thresholds. Unhealthy backends removed from rotation. HTTP, HTTPS, TCP, SSL.
Backend Services	Defines how traffic distributed. Session affinity options. Connection draining. Timeout configuration.

Cloud CDN

Content Delivery Network integrated with Cloud Load Balancing for fast content delivery.

Service / Concept	Description & Key Points
Global Distribution	100+ PoPs worldwide. Cache content close to users. Reduces latency and bandwidth costs. Integrated with Google's global network.
Cache Control	HTTP cache headers respected. Custom TTLs. Cache invalidation. Signed URLs for private content. Cache keys customization.
HTTPS Support	SSL/TLS termination at edge. Automatic certificate management with Google-managed certificates. Support for custom certificates.
Use Cases	Static content delivery, dynamic content acceleration, media streaming, software distribution, gaming assets.

Cloud DNS

Scalable, reliable, and managed authoritative Domain Name System (DNS) service.

Service / Concept	Description & Key Points
Managed Zones	Container for DNS records. Public zones (internet-accessible) or private zones (VPC-internal). DNSSEC support.
Performance	100% SLA. Anycast name servers. Low latency worldwide. Automatic scaling.
Record Types	A, AAAA, CNAME, MX, TXT, SRV, NS, SOA, CAA. Wildcard records. Alias records (special handling).
Private DNS	Internal name resolution within VPC. Split-horizon DNS. No exposure to internet. Integration with on-premises via Cloud VPN/Interconnect.

Cloud VPN

Secure connection between on-premises network and GCP VPC through IPsec VPN.

Service / Concept	Description & Key Points
HA VPN	99.99% SLA. Two interfaces for redundancy. Dynamic routing with Cloud Router. Recommended for production.
Classic VPN	99.9% SLA. Single interface. Static or dynamic routing. Legacy, use HA VPN for new deployments.
Encryption	IPsec tunnel encryption. Secure data in transit. Multiple encryption algorithms supported.

Service / Concept	Description & Key Points
Routing	Static routes (manual configuration) or dynamic routes with BGP via Cloud Router. Automatic failover with HA VPN.

Cloud Interconnect

Dedicated, high-bandwidth connection between on-premises and GCP.

Service / Concept	Description & Key Points
Dedicated Interconnect	Direct physical connection. 10 Gbps or 100 Gbps circuits. Meet Google at colocation facility. Lowest latency, highest throughput.
Partner Interconnect	Connection through service provider. 50 Mbps to 50 Gbps. Don't need to be in Google colocation. More flexible, easier setup.
Benefits	Lower egress costs. Predictable performance. Private connectivity (not over internet). Higher bandwidth than VPN.
Use Cases	Large data transfers, hybrid cloud, disaster recovery, low-latency requirements, regulatory compliance.

Data & Analytics Services

BigQuery

Serverless, highly scalable data warehouse for analytics with built-in ML capabilities.

Service / Concept	Description & Key Points
Serverless	No infrastructure to manage. Automatic scaling. Pay for queries and storage only. Separates compute and storage.
Performance	Petabyte-scale analysis. Columnar storage. Parallel query execution. In-memory BI Engine for sub-second queries.
SQL Interface	Standard SQL (ANSI SQL 2011). User-defined functions (UDF). Stored procedures. Scripting support.
Data Loading	Batch loads from Cloud Storage, streaming inserts. Data Transfer Service for automated loads. Federated queries (query external sources).
Integration	BigQuery ML (create ML models with SQL). Geo support. BI tools integration (Looker, Tableau, Data Studio). Export to Cloud Storage.
Cost Control	On-demand pricing or flat-rate. Partition and cluster tables to reduce costs. Query cost estimation before running.

Dataflow

Fully managed stream and batch data processing based on Apache Beam.

Service / Concept	Description & Key Points
Unified Model	Same code for batch and streaming. Apache Beam SDK. Portable across multiple runners. Python, Java, Go support.
Streaming	Real-time data processing. Exactly-once processing. Windowing and triggers. Watermarks for late data handling.
Batch Processing	Process large datasets. Automatic resource optimization. Dynamic work rebalancing. Fault tolerance.
Auto-scaling	Horizontal and vertical auto-scaling. Right fitting (optimize resources). Streaming engine for improved performance.
Use Cases	ETL pipelines, real-time analytics, data enrichment, log analysis, click stream analysis, IoT data processing.

Dataproc

Managed Apache Spark and Hadoop service for big data processing and machine learning.

Service / Concept	Description & Key Points
Hadoop Ecosystem	Spark, Hadoop, Pig, Hive, Presto. Pre-configured clusters. Custom images support. Initialization actions.
Fast Provisioning	Clusters in 90 seconds. Per-second billing. Ephemeral clusters recommended. Separate storage from compute.
Scaling	Manual or auto-scaling. Scale up/down without job interruption. Preemptible workers for cost savings.
Integration	BigQuery connector. Cloud Storage (HDFS replacement). Dataproc Metastore. Jupyter notebooks. Workflow orchestration.
Use Cases	Migrate existing Spark/Hadoop workloads. Batch processing. Machine learning with Spark ML. Data lake analytics.

Pub/Sub

Real-time messaging service for asynchronous event-driven systems and streaming analytics.

Service / Concept	Description & Key Points
Messaging Pattern	Publishers send messages to topics. Subscribers receive from subscriptions. Many-to-many relationships. Decouples services.
Delivery	At-least-once delivery guarantee. Messages stored until acknowledged. Configurable retention (up to 31 days). Ordering keys for ordered delivery.
Scaling	Automatic scaling. Global service. Handles millions of messages per second. No provisioning needed.
Push and Pull	Pull (subscriber requests messages) or Push (messages sent to webhook endpoint). Dead letter topics for failed messages.
Use Cases	Event streaming, real-time analytics, IoT data ingestion, microservices communication, data replication, log aggregation.
Integration	Dataflow source/sink. Cloud Functions trigger. Cloud Run trigger. BigQuery subscription for direct loading.

Cloud Composer

Managed Apache Airflow for workflow orchestration across clouds and on-premises.

Service / Concept	Description & Key Points
Workflow Orchestration	DAG-based workflows. Python-based definitions. Schedule and monitor data pipelines. Dependency management.

Service / Concept	Description & Key Points
Managed Service	Fully managed Airflow. Automatic updates. Monitoring and logging. High availability across zones.
Integration	Native GCP operators. Multi-cloud support (AWS, Azure). Hybrid environments. BigQuery, Dataflow, Dataproc operators.
Use Cases	ETL/ELT pipelines. ML pipeline orchestration. Data warehouse automation. Multi-system workflows.

AI & Machine Learning Services

Vertex AI

Unified platform for building, deploying, and scaling ML models with MLOps tools.

Service / Concept	Description & Key Points
Unified Platform	End-to-end ML workflow. Data preparation, training, deployment, monitoring. AutoML and custom models. Pre-built containers.
AutoML	No-code ML model training. Vision, video, NLP, tabular data. Automatic feature engineering. Model evaluation and tuning.
Custom Training	Bring your own code. TensorFlow, PyTorch, scikit-learn support. Distributed training. Hyperparameter tuning.
Deployment	Managed endpoints for predictions. Batch predictions. Online and offline serving. Auto-scaling. Model versioning.
MLOps	Vertex AI Pipelines for workflow orchestration. Experiments for tracking. Model monitoring. Feature Store.
Notebooks	Managed JupyterLab. Pre-installed ML frameworks. GPU support. Git integration.

Pre-trained AI APIs

Ready-to-use AI capabilities through REST APIs without ML expertise required.

Service / Concept	Description & Key Points
Vision AI	Image analysis and classification. Object detection. OCR (text extraction). Face detection. Explicit content detection. Logo detection.
Natural Language AI	Entity recognition. Sentiment analysis. Syntax analysis. Content classification. Text translation integration.
Translation AI	100+ languages. Neural machine translation. Custom models. Glossaries for domain-specific terms. Document translation.
Speech-to-Text	Audio transcription. 125+ languages. Real-time and batch. Speaker diarization. Punctuation. Custom vocabulary.
Text-to-Speech	40+ languages. 220+ voices. WaveNet and Neural2 voices. Custom voice. SSML support. Audio profiles.
Video Intelligence	Label detection. Shot change detection. Explicit content detection. Text detection. Object tracking. Logo recognition.

Service / Concept	Description & Key Points
Document AI	OCR for documents. Form parsing. Invoice processing. Custom extractors. Identity document parsing.

Security & Identity Services

Cloud IAM

Fine-grained access control and visibility for centrally managing cloud resources.

Service / Concept	Description & Key Points
Principle of Least Privilege	Grant minimum permissions needed. Use predefined roles when possible. Create custom roles for specific needs. Regular access reviews.
Resource Hierarchy	Permissions inherited from parent to child. Set policies at organization, folder, project, or resource level. Lower level policies can't remove permissions from higher levels.
Conditions	Conditional access based on attributes. Time-based access. IP address restrictions. Resource attributes. VPC Service Controls integration.
Policy Intelligence	Policy Analyzer (understand who has access). Policy Troubleshooter (debug access issues). Recommender (identify over-privileged roles).
Audit Logging	Admin activity logs (who did what). Data access logs (who accessed data). System event logs. Retention up to 400 days.

Secret Manager

Secure and convenient storage system for API keys, passwords, certificates, and other sensitive data.

Service / Concept	Description & Key Points
Secret Storage	Store API keys, passwords, certificates. Version management. Automatic replication. Global or regional storage.
Access Control	IAM integration. Audit logging. VPC Service Controls. Encryption at rest and in transit. Customer-managed encryption keys (CMEK).
Versioning	Multiple versions per secret. Enable/disable versions. Automatic rotation support. Pin to specific versions.
Integration	Cloud Functions, Cloud Run, GKE (via volumes). Cloud Build. Direct API access. Client libraries for multiple languages.

Cloud KMS

Manage encryption keys for cloud services and customer applications.

Service / Concept	Description & Key Points
Key Types	Symmetric (same key encrypt/decrypt). Asymmetric (public/private key pairs). HMAC (signing). Multiple algorithms supported.
Key Hierarchy	Key rings (collection of keys). Keys (multiple versions). Key versions (actual cryptographic material). Rotation policies.
Key Protection	Software keys (FIPS 140-2 Level 1). HSM keys (FIPS 140-2 Level 3). External keys (on-premises HSM). EKM (External Key Manager).
Integration	Customer-Managed Encryption Keys (CMEK) for GCP services. Cloud Storage, Compute Engine, BigQuery. Application-level encryption.
Rotation	Automatic rotation. Manual rotation. Key versions tracked. Old versions retained for decryption.

Security Command Center

Centralized security and risk management platform for GCP resources.

Service / Concept	Description & Key Points
Asset Discovery	Inventory of all GCP resources. Asset changes over time. Security marks for categorization. Export to BigQuery for analysis.
Vulnerability Detection	Web Security Scanner. OS vulnerability detection. Container vulnerability scanning. Misconfiguration detection.
Threat Detection	Event Threat Detection (suspicious activity). Container Threat Detection. VM Threat Detection. Anomaly detection.
Compliance	CIS benchmarks. PCI-DSS. HIPAA. Custom compliance frameworks. Continuous monitoring.

Cloud Armor

DDoS protection and Web Application Firewall (WAF) for applications running on GCP.

Service / Concept	Description & Key Points
DDoS Protection	Layer 3/4 DDoS mitigation. Always-on detection. Automatic mitigation. Global anycast network. Network edge protection.
WAF	OWASP Top 10 protection. SQL injection prevention. XSS protection. Custom rules. Preconfigured WAF rules. Rate limiting.
Security Policies	IP allowlists/denylists. Geo-based filtering. Adaptive protection (ML-based). Preview mode before enforcement.

Service / Concept	Description & Key Points
Integration	Works with Cloud Load Balancing. Logging to Cloud Logging. Monitoring in Cloud Monitoring. Security Command Center integration.

DevOps & Management Tools

Cloud Build

Serverless CI/CD platform for building, testing, and deploying across multiple environments.

Service / Concept	Description & Key Points
Build Configuration	YAML-based build steps. Multiple builders (Docker, Maven, npm, etc.). Custom builders. Parallel execution of steps.
Triggers	GitHub, Bitbucket, Cloud Source Repositories. Webhook triggers. Manual triggers. Pub/Sub triggers. Tag-based triggers.
Security	Private pools (build in your VPC). Vulnerability scanning. Binary authorization. SLSA compliance. Secret Manager integration.
Artifacts	Build artifacts to Cloud Storage or Artifact Registry. Container image builds. Multi-stage builds. Build caching.
Integration	Deploy to GKE, Cloud Run, App Engine, Cloud Functions. GitHub integration. Test result visualization.

Artifact Registry

Universal package management repository for container images and language packages.

Service / Concept	Description & Key Points
Supported Formats	Docker/OCI containers. Maven (Java). npm (Node.js). Python packages. Go modules. Apt (Debian/Ubuntu). Yum (RedHat/CentOS).
Repository Types	Standard (single format). Remote (proxy to external). Virtual (aggregate multiple repos). Multi-region replication.
Security	Vulnerability scanning. IAM integration. VPC Service Controls. Customer-managed encryption. Artifact analysis.
Integration	Cloud Build. GKE. Cloud Run. Compute Engine. Binary Authorization. Cloud Deploy.

Cloud Logging

Real-time log management and analysis for applications and infrastructure.

Service / Concept	Description & Key Points
Log Types	Admin Activity (who did what). Data Access (who accessed what). System Events (GCP system changes). User-generated logs.

Service / Concept	Description & Key Points
Log Router	Route logs to destinations. Include/exclude filters. Multiple sinks per log. Route to Cloud Storage, BigQuery, Pub/Sub, or other projects.
Log Retention	Admin Activity: 400 days. Data Access: 30 days (configurable). Custom retention with log buckets. Archive to Cloud Storage.
Log Analysis	Logs Explorer (query interface). Log-based metrics. Integration with Cloud Monitoring. Export to BigQuery for analysis.
Compliance	Audit logs for compliance. Immutable logs. Access transparency logs (Google employee access). Locked log buckets.

Cloud Monitoring

Visibility into performance, uptime, and health of cloud-powered applications.

Service / Concept	Description & Key Points
Metrics	System metrics (CPU, memory, disk). Service metrics (latency, errors). Custom metrics. OpenTelemetry support. Prometheus integration.
Dashboards	Pre-built dashboards. Custom dashboards. Charts and visualizations. Multiple metrics per chart. Sharing and exporting.
Alerting	Metric-based alerts. Log-based alerts. Uptime checks. Notification channels (email, SMS, Pub/Sub, webhooks, PagerDuty). Alert policies.
Uptime Monitoring	HTTP/HTTPS checks. TCP checks. Custom checks. Global monitoring locations. SSL certificate expiration monitoring.
SLO Monitoring	Define Service Level Objectives. Track error budgets. Burn rate alerts. Compliance tracking.

Cloud Deployment Manager

Infrastructure as Code service for automating resource creation and management.

Service / Concept	Description & Key Points
Templates	YAML or Python templates. Parameterized configurations. Reusable templates. Template imports.
Deployments	Declarative resource management. Update, delete, or preview deployments. Rollback capability. Dependency management.
Use Cases	Reproducible environments. Multi-environment deployments. Infrastructure versioning. Complex multi-resource setups.

Service / Concept	Description & Key Points
Alternative	Terraform is often preferred for multi-cloud. Deployment Manager is GCP-native. Both are Infrastructure as Code.

Hybrid & Multi-Cloud

Anthos

Application platform for hybrid and multi-cloud environments with consistent management.

Service / Concept	Description & Key Points
Platform Overview	Kubernetes-based platform. Run anywhere (GCP, on-prem, other clouds). Consistent management layer. Service mesh (Anthos Service Mesh).
GKE Enterprise	Formerly Anthos. Managed Kubernetes across environments. Unified policy and security. Multi-cluster management. Config management.
Anthos Clusters	GKE on AWS, Azure, bare metal, VMware. Consistent APIs and tooling. Centralized configuration. Fleet management.
Service Mesh	Based on Istio. Traffic management. Security (mTLS). Observability. Multi-cluster service discovery.
Config Management	GitOps-based configuration. Policy enforcement. Multi-cluster configuration sync. Hierarchical policies.
Use Cases	Modernize on-premises apps. Multi-cloud strategy. Edge computing. Regulatory requirements. Vendor diversification.

Cost Management & Optimization

Pricing Models

Service / Concept	Description & Key Points
On-Demand Pricing	Pay-as-you-go. No upfront costs. Per-second billing for Compute Engine. Flexible scaling.
Sustained Use Discounts	Automatic discounts for running resources >25% of month. Up to 30% discount. No commitment required. Applied automatically.
Committed Use Discounts	1 or 3 year commitment. Up to 57% discount. Flexible or resource-based commitments. Covers specific machine types or spend amount.
Preemptible/Spot VMs	Up to 91% discount. Can be terminated anytime. 24-hour max runtime (preemptible). No max runtime (spot). For fault-tolerant workloads.
Free Tier	Always free products (limited usage). 90-day \$300 credit for new customers. Compute Engine f1-micro, Cloud Storage, BigQuery queries.

Cost Optimization Strategies

Service / Concept	Description & Key Points
Right-Sizing	Use appropriate machine types. Resize underutilized VMs. Custom machine types for exact fit. Active Assist recommendations.
Auto-Scaling	Scale resources based on demand. Managed instance groups. GKE cluster autoscaler. Cloud Run concurrency.
Storage Optimization	Use appropriate storage classes. Lifecycle policies for Cloud Storage. Delete unused snapshots/images. Nearline/Coldline for archival.
Network Optimization	Use internal IPs when possible. Cloud CDN for content delivery. Regional resources to reduce cross-region traffic. VPC peering vs VPN.
Monitoring & Budgets	Set budget alerts. Use cost breakdown by project/label. Billing export to BigQuery. Active Assist cost recommendations.

GCP Best Practices

Security Best Practices

- Use service accounts with minimal necessary permissions (principle of least privilege)
- Enable Cloud Audit Logs for all projects and critical resources
- Implement VPC Service Controls for sensitive data protection

- Use Secret Manager instead of hardcoding credentials
- Enable encryption at rest and in transit (default for most services)
- Implement multi-factor authentication (MFA) for all users
- Use VPC firewall rules to restrict access to resources
- Regularly review IAM permissions and remove unused access
- Use Cloud KMS for customer-managed encryption keys (CMEK)
- Enable Security Command Center for centralized security monitoring

Architecture Best Practices

- Design for high availability using multi-zonal or multi-regional deployments
- Implement disaster recovery plans with regular backup testing
- Use managed services when possible to reduce operational overhead
- Decouple services using Pub/Sub for event-driven architectures
- Implement caching strategies (Memorystore, Cloud CDN) to reduce latency
- Use health checks and auto-healing for resilient applications
- Design stateless applications for easier scaling
- Use load balancing for traffic distribution and failover
- Implement proper network segmentation with VPCs and subnets
- Choose appropriate database based on use case (SQL, NoSQL, analytical)

Operational Best Practices

- Implement comprehensive monitoring and alerting (Cloud Monitoring)
- Use labels and tags consistently for resource organization and cost tracking
- Set up budget alerts to monitor and control spending
- Automate infrastructure provisioning with Terraform or Deployment Manager
- Implement CI/CD pipelines with Cloud Build for consistent deployments
- Use separate projects for different environments (dev, staging, production)
- Centralize logging and export to BigQuery for long-term analysis
- Document architecture decisions and maintain runbooks
- Perform regular security audits and penetration testing
- Keep services and dependencies updated with latest patches

Performance Optimization

- Choose regions close to users for lower latency
- Use Cloud CDN for static content and media delivery

- Implement connection pooling for database connections
- Use appropriate machine types and storage classes for workload
- Leverage BigQuery partitioning and clustering for better query performance
- Use persistent disk snapshots for faster VM creation
- Optimize container images for faster startup times
- Enable HTTP/2 and connection multiplexing for better throughput
- Use premium network tier for performance-critical applications
- Profile applications and optimize based on metrics